

# Levi Harris

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## RESEARCH INTERESTS

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Remote Sensing, Hydrometeorology, Deep Learning

## EDUCATION

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### Northeastern University

Ph.D. Interdisciplinary Engineering – *Advisor: Auroop Ganguly*

Boston, MA

*Fall 2026-Present*

- Primary Department: Civil & Environmental Engineering
- *Sami Alsaif Doctoral Fellow*

### University of North Carolina at Chapel Hill

M.S. Computer Science – *Advisor: Tianlong Chen*

B.S. Computer Science

Chapel Hill, NC

*Fall 2025*

*Spring 2024*

- Minor: Data Science

## EXPERIENCE

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### University of Texas at Austin

Research Scientist I, *Jackson School of Geosciences, Advisor: Dev Niyogi*

Austin, TX

*Summer 2026*

### University of North Carolina at Chapel Hill

GRA, *Department of Computer Science, Advisor: Tianlong Chen*

URA, *Department of Computer Science, Advisor: Gedas Bertasius*

Chapel Hill, NC

*Fall 2024-Winter 2025*

*Spring 2023-Fall 2024*

### The National Weather Service

Summer volunteer, *Mentor: Stanley Czyzyk*

Las Vegas, NV

*Summer 2025*

## PUBLICATIONS

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- 1 Levi Harris, Tianlong Chen, “A Space-Time Transformer for Precipitation Nowcasting”, **1st Place NeurIPS 2025 Weather4Cast Challenge**, *NeurIPS 2025 Weather4Cast Challenge Workshop; arXiv:2511.11090*, 2025.
- 2 Levi Harris, Md Jayed Hossain, Mufan Qiu, Ruichen Zhang, Pingchuan Ma, Tianlong Chen, Jiaqi Gu, Seth Ariel Tongay, Umberto Celano, “SparseC-AFM: a deep learning method for fast and accurate characterization of MoS<sub>2</sub> with C-AFM,” *SPIE Conference on Low-Dimensional Materials and Devices; paper 13582-35*, 2025.

## PREPRINTS

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- 1 Oscar Ovanger, Levi Harris, Timothy H. Keitt, “Adaptive Evidence Weighting for Audio-Spatiotemporal Fusion”, *arXiv preprint arXiv:2602.03817*, 2026.
- 2 Levi Harris, Dev Niyogi “MRMS Wet Biases Enhanced by Sub-Cloud Evaporation in the Desert Southwest”, 2026. *In Prep. Journal of Applied Meteorology and Climatology*
- 3 Levi Harris, Auroop Ganguly, “Decomposing Sources of MRMS QPE Error with Deep Mixture-of-Expert Models”, 2026. *In Preparation for the Journal of Hydrometeorology*

## TALKS/PANEL DISCUSSIONS

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| 1. “ <b>SaTformer</b> ” [Oral]<br><i>NeurIPS 2025 Weather4Cast Challenge</i>                               | 6 December 2025<br>San Diego, CA    |
| 2. “ <b>SparseC-AFM</b> ” [Author]<br><i>SPIE Conference on Low-Dimensional Materials and Devices 2025</i> | 17 September 2025<br>San Diego, CA  |
| 3. “ <b>Decoding Graduate Research at UNC</b> ” [Panelist]<br><i>UNC CS for Social Good</i>                | 14 October 2025<br>Chapel Hill, NC  |
| 4. “ <b>Life as an undergraduate researcher</b> ”<br><i>UNC AI Club Symposium</i>                          | 29 November 2023<br>Chapel Hill, NC |

## AWARDS

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### Northeastern University

- Sami Alsaif Doctoral Fellowship 2026

### University of North Carolina at Chapel Hill

- Department Tuition Award 2024, 2025
- Undergraduate Research Stipend 2023, 2024
- Dean’s List 2022, 2023, 2024

## TEACHING

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### DATA 120: Ethics of Artificial Intelligence

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|-----------------------------------------------------------|--------------------|
| <i>Teaching Assistant — Instructor: Santiago Olivella</i> | <i>Fall 2025</i>   |
| <i>Teaching Assistant — Instructor: Justin Sola</i>       | <i>Spring 2025</i> |
| <i>Teaching Assistant — Instructor: Neil Gaikwad</i>      | <i>Fall 2024</i>   |

## MENTORING

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- **Asyat Sow** (UNC Chancellor’s Scholar) *Summer 2025*  
Undergraduate · Project Mentor  
“*Failure Modes of Real-Time Speech-to-Speech Translation Systems on French Regional Dialects*”
- **Wonjae Oh** *Fall 2024*  
Undergraduate · Project Mentor  
“*A Day-One Jailbreak for GPT-4V using ASCII Ciphers*”

## TECHNICAL SKILLS

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|                              |                       |
|------------------------------|-----------------------|
| <b>Programming Languages</b> | Python (fluent), Java |
| <b>Libraries</b>             | PyTorch, TensorFlow   |

## SELECTED COURSES

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Applied Probability and Random Processes · Multivariate Calculus · Introduction to Probability · Linear Algebra · Introduction to Machine Learning · Computational Imaging · Video Recognition with Transformers · Computer Vision of our 3D World · Efficient Machine Learning · Transfer Learning

## REFERENCES

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### Auroop R. Ganguly

*Professor, Civil and Environmental Engineering, Northeastern University*

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Website

**Edward Beighley**

*Professor and Chair, Civil and Environmental Engineering, Northeastern University*  
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**Dev Niyogi**

*Professor, Department of Civil, Architectural, and Environmental Engineering, The University of Texas at Austin*  
E-mail: dev.niyogi@jsg.utexas.edu  
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**Tianlong Chen**

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